

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

```
np.random.seed(42)
```

```
X = 2 * np.random.rand(100, 1)
y = 4 + 3 * X + np.random.randn(100, 1)

df = pd.DataFrame({"X": X.flatten(), "y": y.flatten() })
```

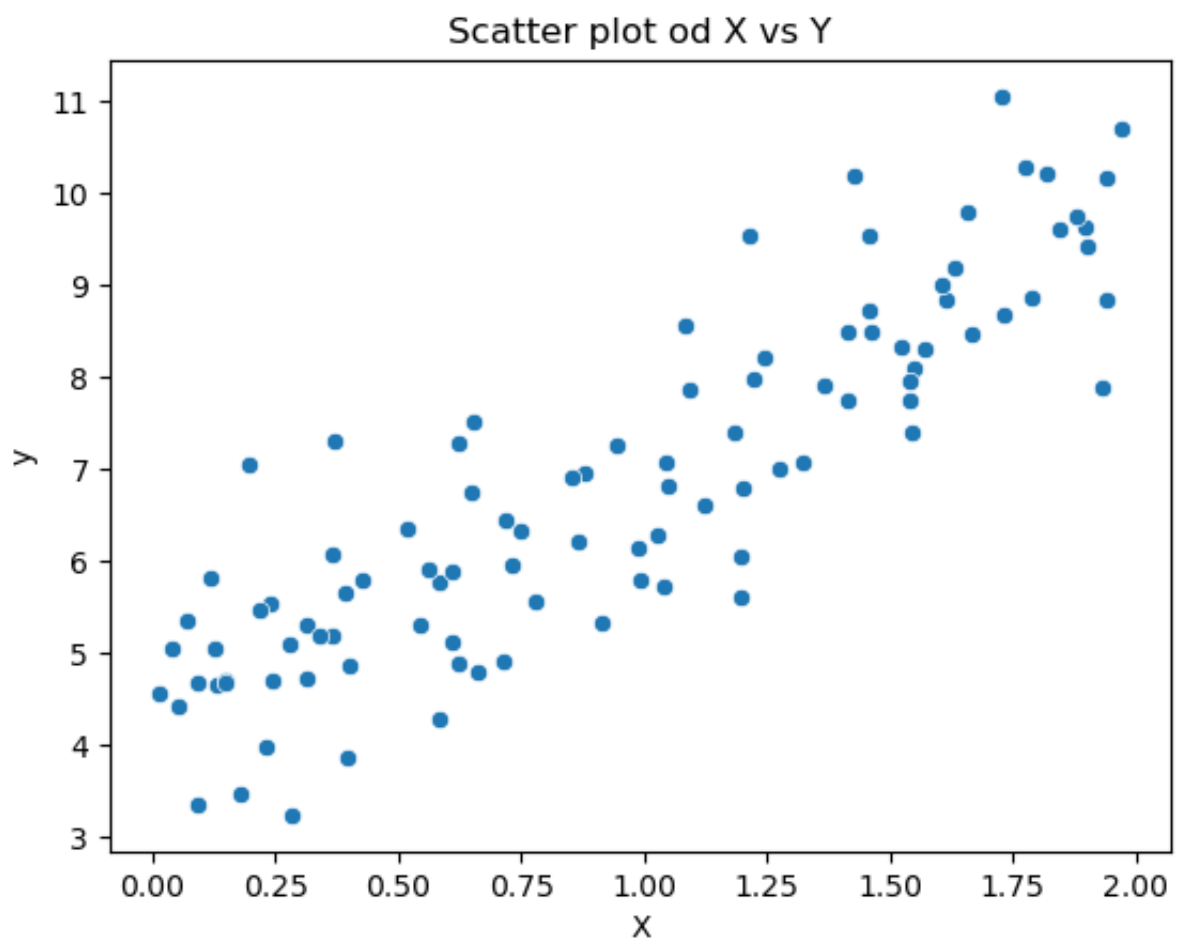
```
df.head()
```

	x	y
0	0.749080	6.334288
1	1.901429	9.405278
2	1.463988	8.483724
3	1.197317	5.604382
4	0.312037	4.716440

```
df.describe()
```

	x	y
count	100.000000	100.000000
mean	0.940361	6.820004
std	0.594979	1.879138
min	0.011044	3.238062
25%	0.386402	5.302794
50%	0.928285	6.676056
75%	1.460406	8.297638
max	1.973774	11.044395

```
sns.scatterplot(x="X", y="y", data = df)
plt.title("Scatter plot od X vs Y")
plt.show()
```



```
X_train, X_test, y_train, y_test = train_test_split(X, y,
                                                    test_size = 0.2, random_state = 42)
```

```
def compute_cost(X,y,w,b):
    n = len(y)
    y_pred = w * X + b
    cost = (1 / n) * np.sum((y_pred - y) ** 2)
    return cost
```

```
def compute_gradients(X,y,w,b):
    n = len(y)
    y_pred = w * X + b

    dw = (2 / n) * np.sum((y_pred - y) * X)
    db = (2 / n) * np.sum(y_pred - y)

    return dw, db
```

```
def gradient_descent(X, y, w, b, lr=0.01, epochs = 1000):
    cost_history = []

    for _ in range(epochs):
        dw, db = compute_gradients(X, y, w, b)
        w -= lr * dw
        b -= lr * db

        cost = compute_cost(X, y, w, b)
        cost_history.append(cost)

    return w, b, cost_history
```

```
w_init = 0
b_init = 0

w,b,cost_history = gradient_descent(
    X_train.flatten(),
    y_train.flatten(),
    w_init,
    b_init,
    lr=0.01,
    epochs=1000
)

print("Learned w:", w)
print("Learned b:", b)
```

```
Learned w: 2.829058820123641
Learned b: 4.1095554202449875
```

```
y_pred_custom = w * X_test + b
y_pred_custom
```

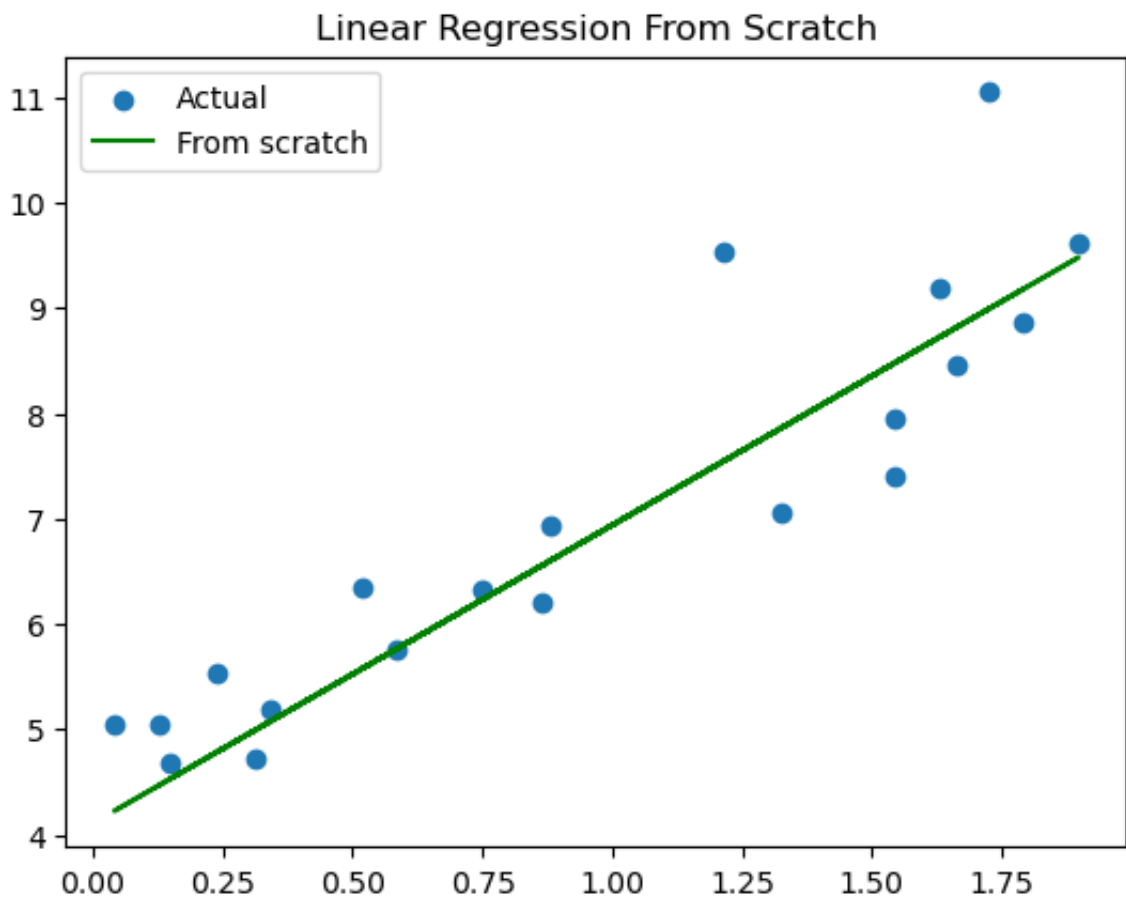
```
array([[4.46917604],
       [9.17259384],
       [8.47900717],
       [7.85818444],
       [5.573763  ],
       [6.59999001],
       [5.76254421],
       [8.99309614],
       [4.22602491],
       [6.22874747],
       [6.55355115],
       [7.54711566],
       [8.72353211],
       [9.47846142],
       [4.78623373],
       [4.99232724],
       [8.47349377],
       [4.52850877],
       [8.81961381],
       [5.07440097]])
```

```
mse_custom = mean_squared_error(y_test, y_pred_custom)
r2_custom = r2_score(y_test, y_pred_custom)

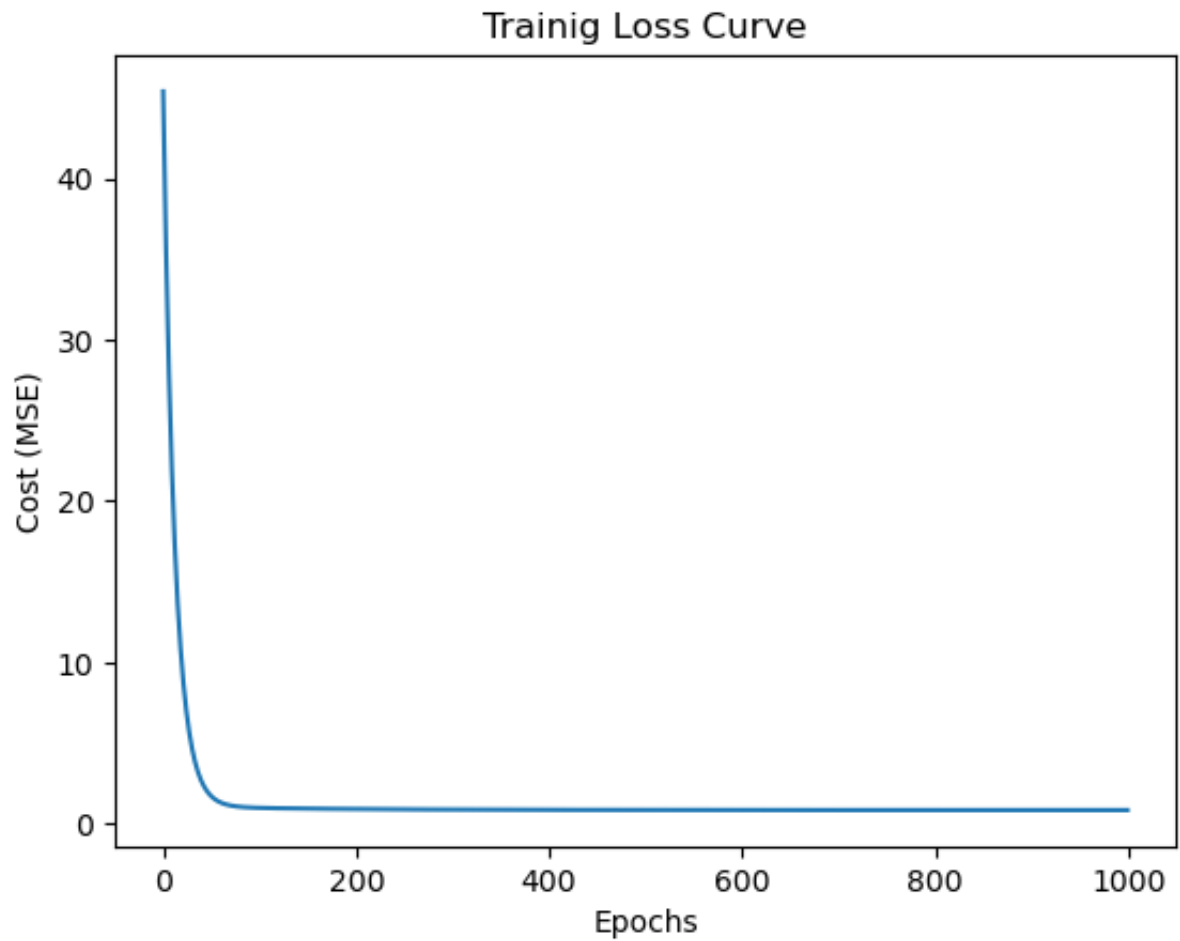
print("Custom MSE:", mse_custom)
print("Custom R2:", r2_custom)
```

```
Custom MSE: 0.6595295175508236
Custom R2: 0.8054865345109443
```

```
plt.scatter(X_test, y_test, label = "Actual")
plt.plot(X_test, y_pred_custom, color="green", label="From scratch")
plt.legend()
plt.title("Linear Regression From Scratch")
plt.show()
```



```
plt.plot(cost_history)
plt.xlabel("Epochs")
plt.ylabel("Cost (MSE)")
plt.title("Trainig Loss Curve")
plt.show()
```



#cost is decreasing hence we know that gradient descent is working

